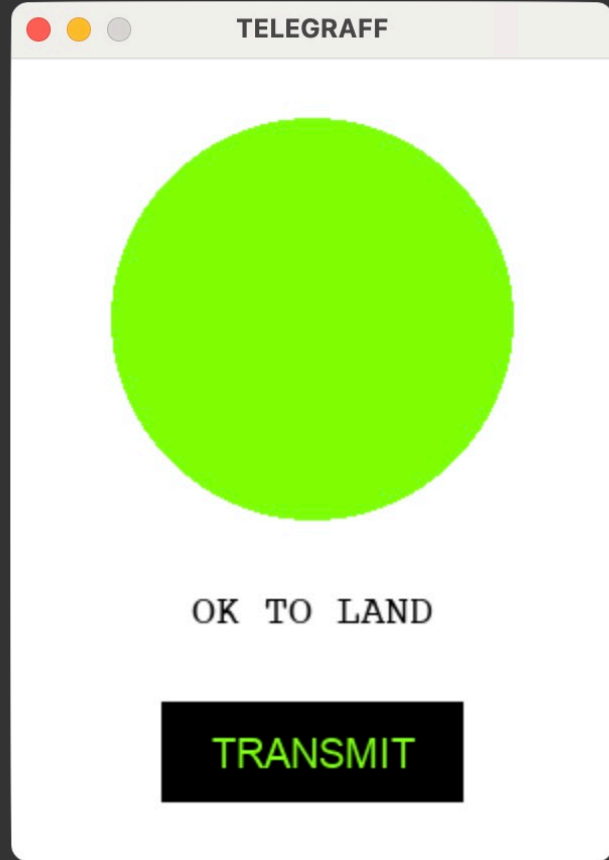


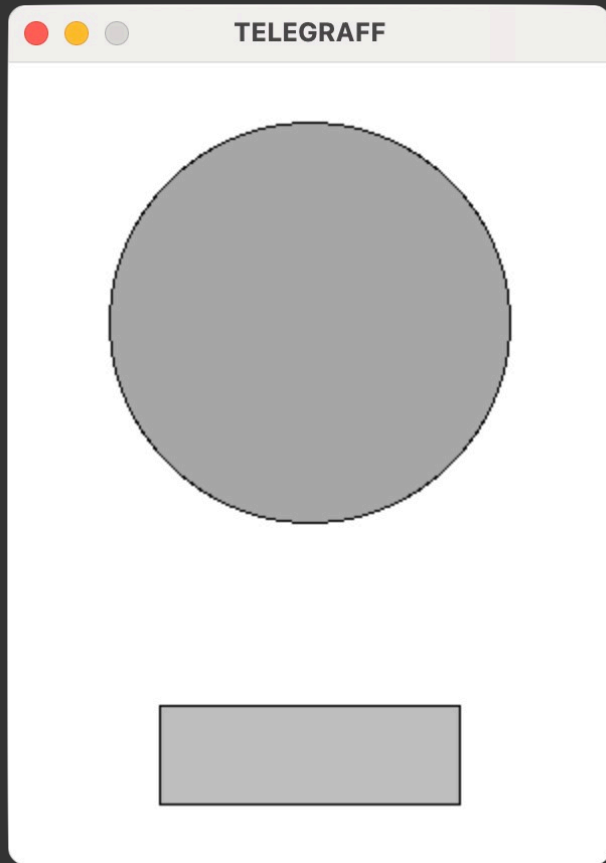
simple  
graphics

CSCI  
134

let's make an app  
that transmits  
morse code

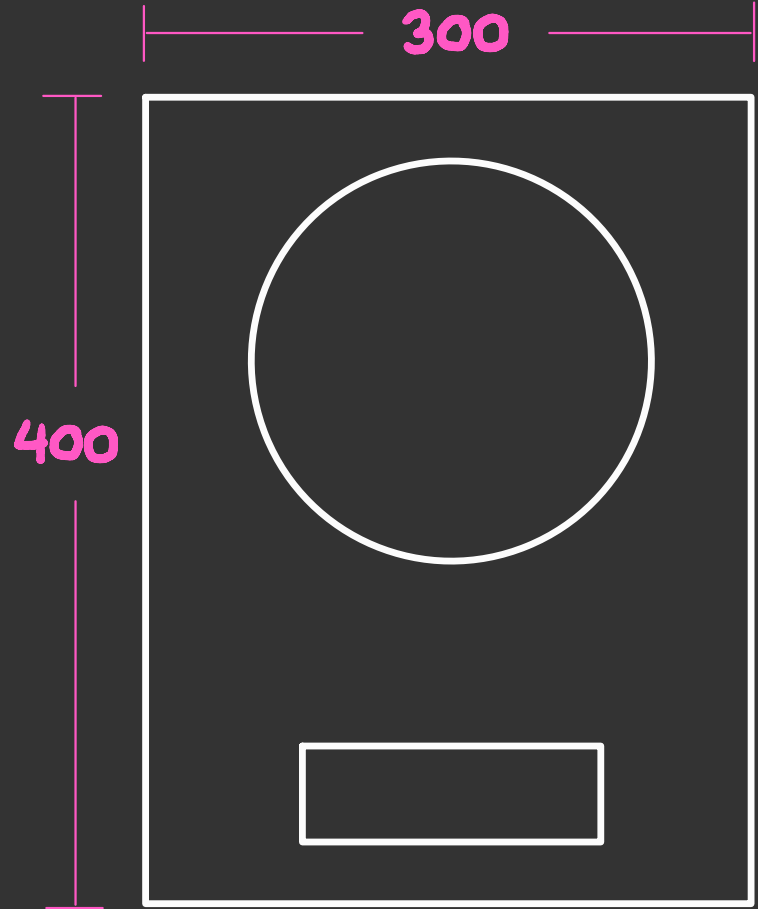


we'll start  
by making a  
"mock up"



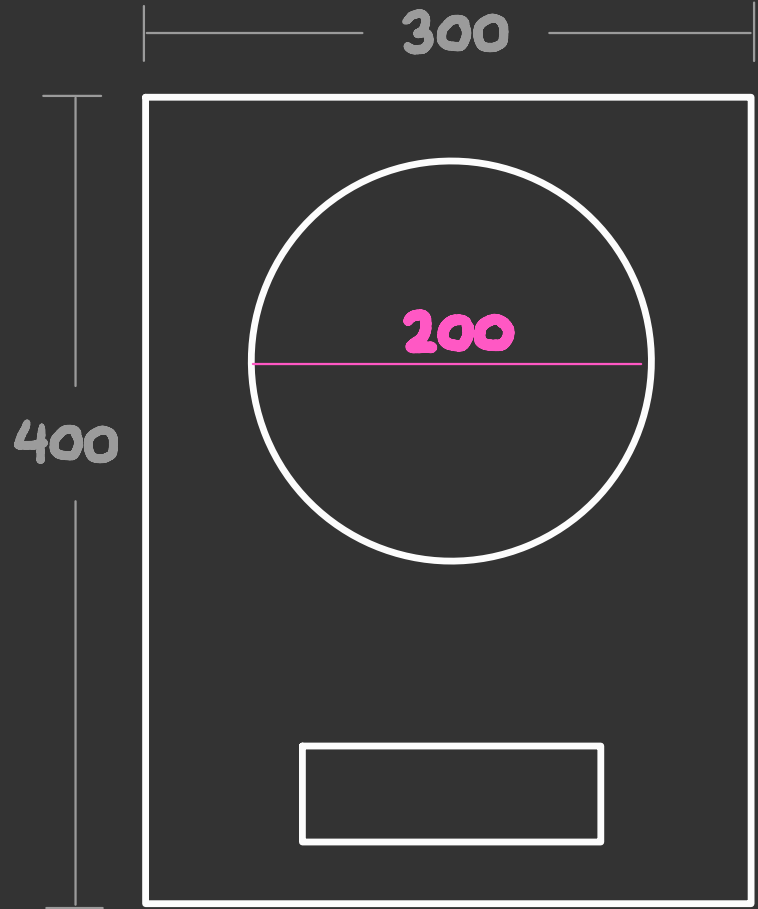
```
from graphics import Window
from graphics import Rectangle, Circle
```

```
window = Window(width=300,
                height=400,
                title="TELEGRAFF")
lightbulb = Circle(diameter=200,
                  fill_color="darkgray",
                  outline_color="black")
button = Rectangle(width=150, height=50,
                  fill_color="gray",
                  outline_color="black")
window.paste(lightbulb, 50, 30)
window.paste(button, 75, 320)
```



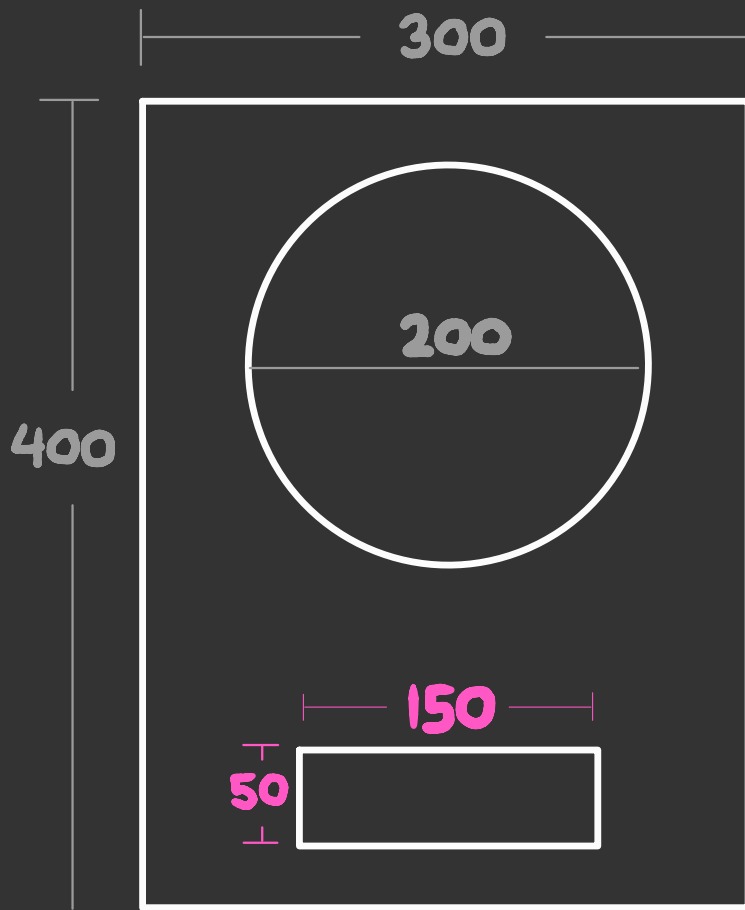
```
from graphics import Window
from graphics import Rectangle, Circle

window = Window(width=300,
                height=400,
                title="TELEGRAFF")
lightbulb = Circle(diameter=200,
                  fill_color="darkgray",
                  outline_color="black")
button = Rectangle(width=150, height=50,
                  fill_color="gray",
                  outline_color="black")
window.paste(lightbulb, 50, 30)
window.paste(button, 75, 320)
```



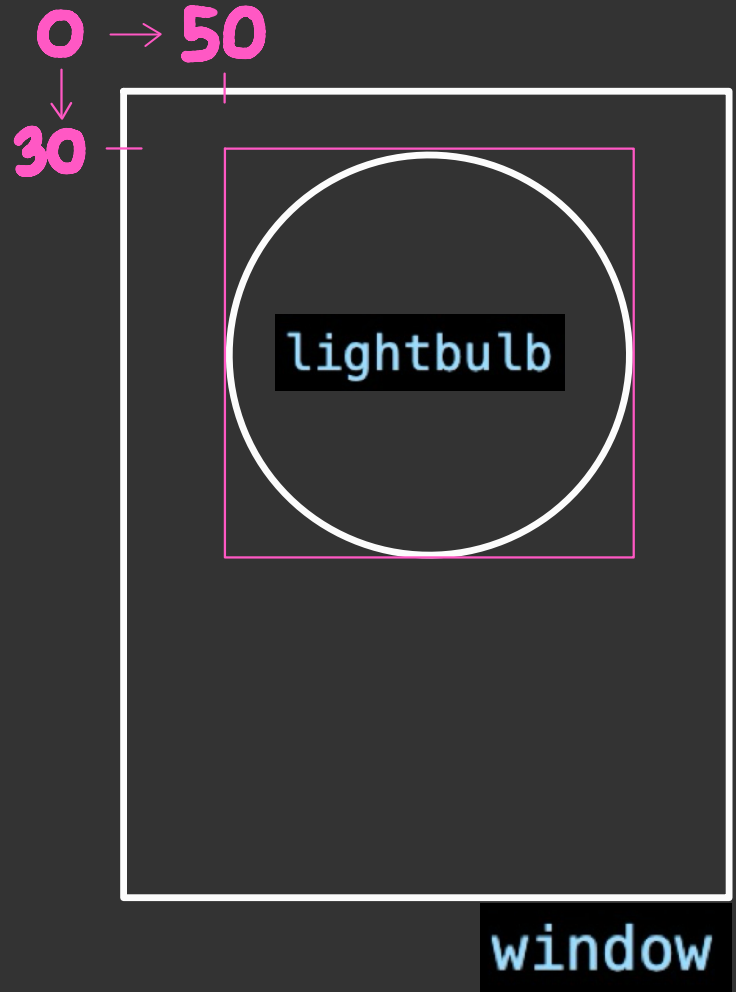
```
from graphics import Window
from graphics import Rectangle, Circle

window = Window(width=300,
                height=400,
                title="TELEGRAFF")
lightbulb = Circle(diameter=200,
                  fill_color="darkgray",
                  outline_color="black")
button = Rectangle(width=150, height=50,
                  fill_color="gray",
                  outline_color="black")
window.paste(lightbulb, 50, 30)
window.paste(button, 75, 320)
```



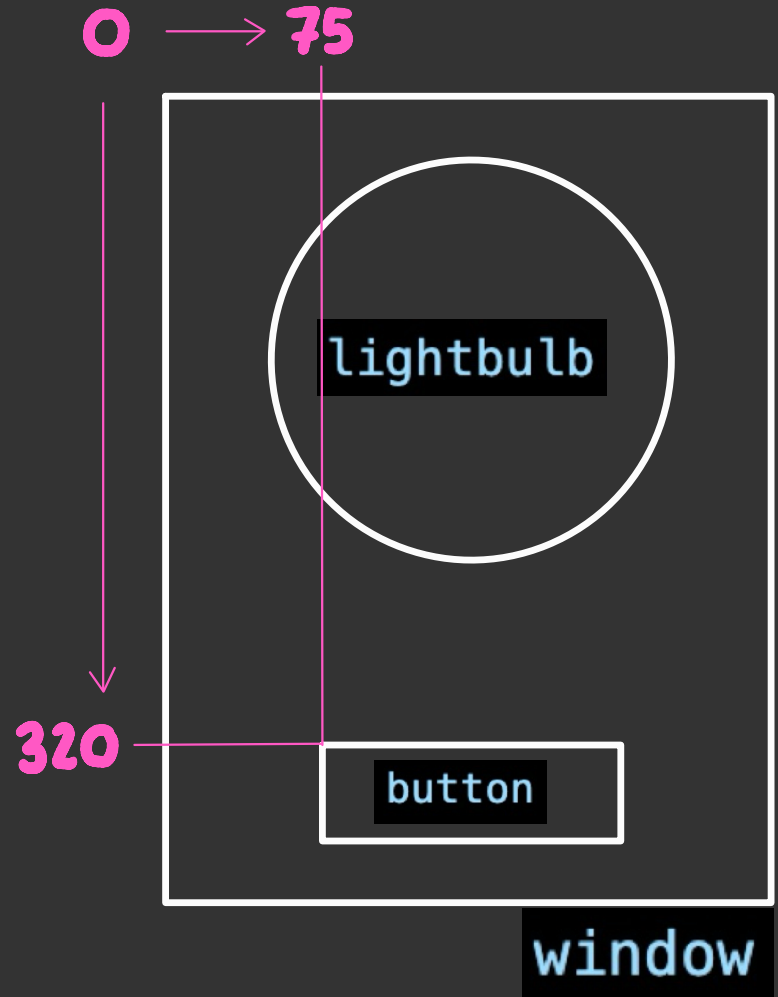
```
from graphics import Window
from graphics import Rectangle, Circle

window = Window(width=300,
                height=400,
                title="TELEGRAFF")
lightbulb = Circle(diameter=200,
                  fill_color="darkgray",
                  outline_color="black")
button = Rectangle(width=150, height=50,
                  fill_color="gray",
                  outline_color="black")
window.paste(lightbulb, 50, 30)
window.paste(button, 75, 320)
```



```
from graphics import Window
from graphics import Rectangle, Circle

window = Window(width=300,
                height=400,
                title="TELEGRAFF")
lightbulb = Circle(diameter=200,
                  fill_color="darkgray",
                  outline_color="black")
button = Rectangle(width=150, height=50,
                  fill_color="gray",
                  outline_color="black")
window.paste(lightbulb, 50, 30)
window.paste(button, 75, 320)
```



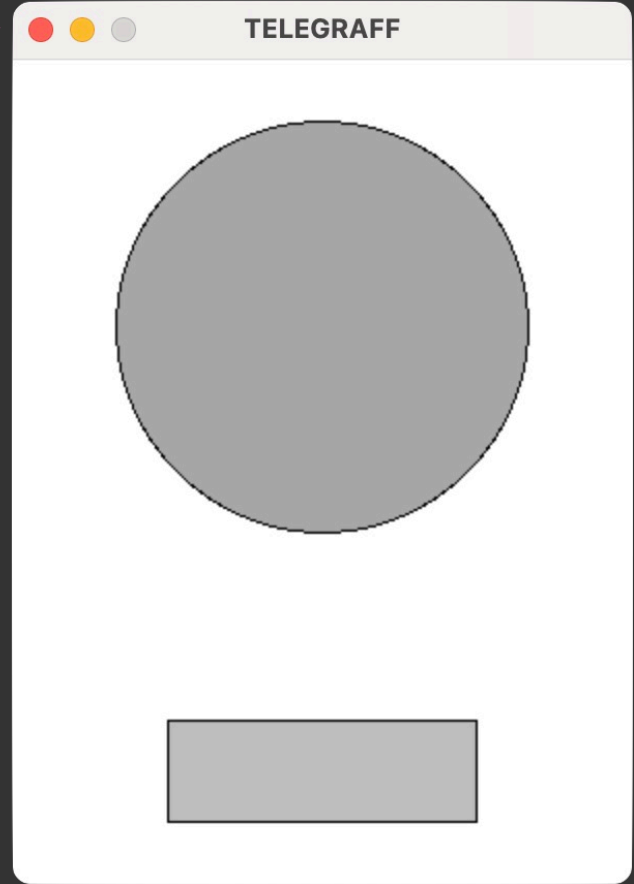


```
from graphics import Window
from graphics import Rectangle, Circle

window = Window(width=300,
                height=400,
                title="TELEGRAFF")
lightbulb = Circle(diameter=200,
                  fill_color="darkgray",
                  outline_color="black")
button = Rectangle(width=150, height=50,
                  fill_color="gray",
                  outline_color="black")
window.paste(lightbulb, 50, 30)
window.paste(button, 75, 320)
```

```
quit_now = False
while not quit_now:
    quit_now = window.refresh()
```

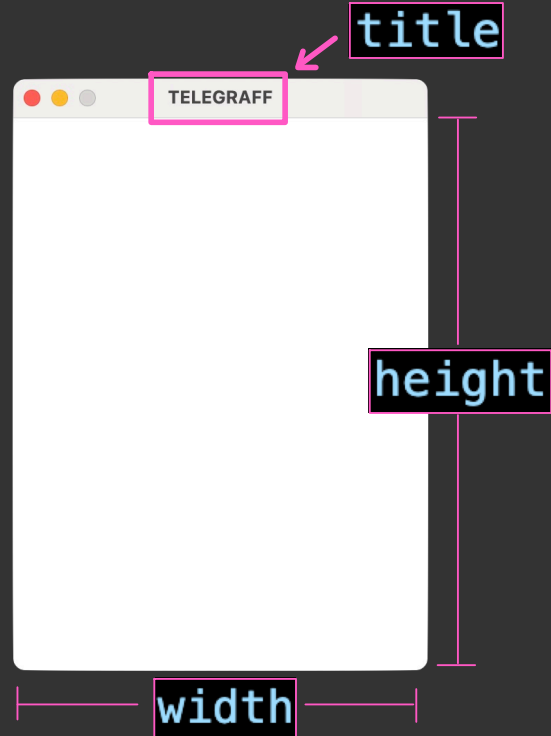
keep  
redrawing  
until  
this  
button  
is pressed



# the Window class

```
window = Window(width, height, title)
```

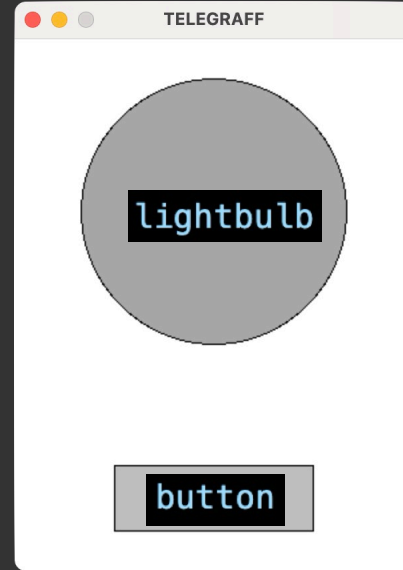
creates an empty  
window



# the Window class

```
window.paste(graphic, x, y)
```

pastes a graphic object (e.g. Circle, Rectangle) at the specified coordinates

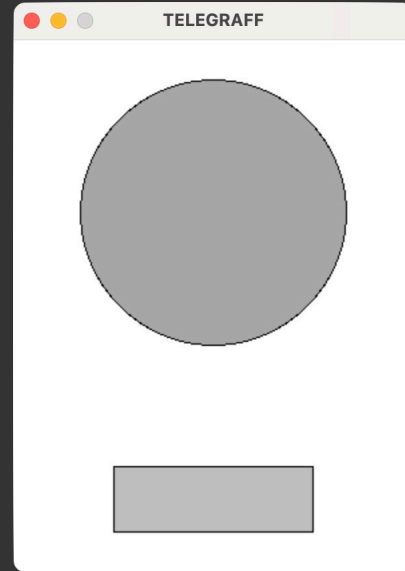


```
window.paste(lightbulb, 50, 30)  
window.paste(button, 75, 320)
```

# the Window class

```
window.refresh()
```

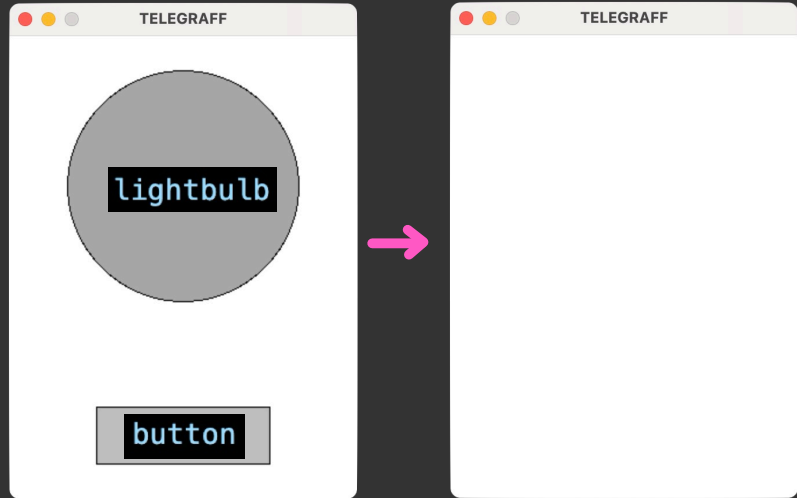
redraws the window  
(pasted elements do not  
appear until this  
method is called)



# the Window class

```
window.clear()
```

clears the  
window

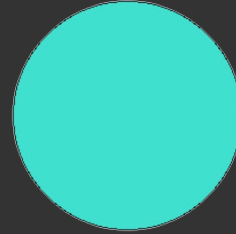


# graphic objects

```
Rectangle(width, height,  
| | | fill_color, outline_color)
```



```
Circle(diameter, fill_color, outline_color)
```



```
Textbox(width, height, msg, fill_color,  
| | text_color, font_name, font_size)
```

